

SemEval-2026 Task 4: Narrative Story Similarity and Narrative Representation Learning Base-Model Implementation

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Abstract

This report details our experimentation for SemEval-2026 Task 4, focusing on Track A: Triplet-based Relative Similarity. We evaluate the official baselines (Random Guess and GPT-4o-mini) and implement two embedding-based approaches with rich feature engineering: a Doc2Vec model combined with multiple classifiers, and a pre-trained Sentence Transformer (all-mpnet-base-v2) with an MLP neural network. For the Doc2Vec approach, we trained on 1,039 examples (39 sample + 1,000 synthetic) and created 44 hand-crafted features from the embeddings, testing five different classifiers including Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and MLP. While cross-validation on the training set showed strong performance (96-97% accuracy), evaluation on the held-out development set revealed significant overfitting, with test accuracies ranging from 48-55%. The Sentence Transformer approach, trained on 539 examples with 49 features, achieved better generalization with 63% test accuracy, though still exhibiting a substantial gap from its 96.66% cross-validation score. Our experiments highlight the challenges of training on synthetic data for narrative similarity tasks, where models achieve high performance on training distributions but struggle to generalize to human-annotated test data. The GPT-4o-mini baseline performed at 49.5%, slightly below random chance, suggesting that zero-shot prompting is insufficient for this comparative narrative similarity task.

1 Introduction

Understanding narrative similarity is a complex challenge that requires models to move beyond surface-level keyword matching and grasp deeper structures like plot, character arcs, and causality. SemEval-2026 Task 4 addresses this challenge through two complementary tracks designed to evaluate different aspects of narrative understanding.

Track A: Triplet-based Relative Similarity. This track frames narrative similarity as a binary classification task. Given a triplet of narratives—an *anchor* story (x) and two candidate stories (a and b)—the system must determine which candidate is narratively closer to the anchor. This formulation tests a model’s ability to make fine-grained comparative judgments about narrative content, requiring

systems to identify which of two alternatives better matches the thematic, structural, or semantic properties of a reference story.

Track B: Narrative Representation Learning. This track focuses on unsupervised representation learning. The goal is to produce a single vector embedding for any given story such that the cosine similarity between story embeddings aligns with human judgments of narrative similarity. This approach evaluates whether models can learn meaningful semantic representations of narratives without direct supervision on similarity comparisons.

Baselines and Our Approach. The task organizers provided baseline systems for both tracks. For Track A, they implemented a Random Guess baseline (establishing the floor performance at approximately 50%) and a GPT-4o-mini prompting approach. For Track B, they provided a Sentence-BERT model (all-MiniLM-L6-v2) baseline.

In this report, we focus exclusively on Track A. We replicate the official baselines and introduce two embedding-based models with rich feature engineering:

1. A **Doc2Vec** model that learns dense vector representations for each story through unsupervised training on the corpus, combined with multiple classifiers (Logistic Regression, Random Forest, Gradient Boosting, MLP, and XGBoost) trained on 44 hand-crafted features derived from the embeddings.
2. A **Sentence Transformer** approach using the pre-trained all-mpnet-base-v2 model to encode stories, combined with an MLP neural network classifier trained on 49 similarity-based features.

Both approaches leverage rich feature engineering that captures various aspects of similarity between story embeddings, including cosine similarities, distance metrics, differential features, and statistical properties. We analyze the performance of these models on synthetic training data and human-annotated development data, examining the challenges of generalization in narrative similarity tasks and the implications of training on synthetic versus human-annotated examples.

2 Data Preparation

2.1 Official Datasets

The organizers provided several datasets in JSONlines format, where each line represents a single JSON object. All story summaries in the datasets are sourced from Wikipedia, ensuring consistency in writing style and content structure.

- **Sample Data:** 39 labeled triplets provided for initial validation and system testing. These samples are also available as individual items to simulate Track B evaluation.
- **Development Data:** 200 labeled triplets intended for model tuning and hyperparameter optimization. Like the sample data, these are also available in individual story format.
- **Synthetic Training Data:** 1,900 triplets generated using large language models. This synthetic data was provided to lower the barrier to entry by making it easier to fine-tune models without requiring extensive human-annotated data. The organizers provided this data in two formats:
 - *Synthetic data for classification:* Formatted for Track A with triplets containing `anchor_text`, `text_a`, `text_b`, and a boolean `text_a_is_closer` label.
 - *Synthetic data for contrastive learning:* Formatted with `anchor_story`, `similar_story`, and `dissimilar_story` fields.

Both formats contain the same examples with an additional `model_name` field indicating which LLM generated each story.

- **Test Data:** A held-out set of 400 triplets plus 849 individual stories for final evaluation. Labels for this dataset will only be released after completion of the shared task.

For our experiments, we focus primarily on the 1,900-sample *Synthetic data for classification* format.

2.2 Data Augmentation via Back-Translation

To increase both the size and diversity of the training data, we performed data augmentation on the 1,900-row synthetic dataset. Our augmentation strategy focused on creating paraphrases of the anchor text, as the anchor serves as the central point of comparison in each triplet.

Our back-translation procedure operated as follows. For each original triplet:

1. We retained the original row in its complete form.
2. We translated the `anchor_text` into three languages: Spanish, French, and German using a machine translation model.

3. We translated each of these foreign-language versions back into English.
4. We created three new training samples, each pairing one back-translated anchor with the *original* `text_a`, `text_b`, and `text_a_is_closer` label.

This process effectively quadrupled our training set, expanding the original 1,900 rows to a new corpus of **7,591 samples**. The rationale behind this approach is that back-translation introduces surface-level paraphrasing while preserving the core semantic content of the narrative. By training on these varied expressions of the same underlying story, we hypothesized that models would learn more robust representations that focus on narrative content rather than specific word choices or phrasing patterns.

3 Track A: Triplet-based Relative Similarity

3.1 Official Baselines

The organizers provided two baseline systems for Track A to establish performance benchmarks:

- **Random Guess:** This baseline randomly predicts whether text A or text B is closer to the anchor with 50% probability for each option. This establishes the floor performance at approximately 50% accuracy, which any meaningful model should exceed.
- **GPT-4o-mini:** A prompt-based approach that queries the OpenAI API with carefully designed prompts to make similarity judgments. This baseline leverages the reasoning capabilities of large language models to make comparative narrative similarity assessments. On the development set of 200 examples, this baseline achieved an accuracy of **49.5%**, which is surprisingly below the random baseline. This suggests that the prompt-based approach struggled with the nuanced comparative judgments required for this task, or that the synthetic data contains patterns that differ from the narratives GPT-4o-mini was trained on.

We implemented two additional models to compare against these baselines, each taking a different approach to the triplet similarity task.

3.2 Doc2Vec with Multiple Classifiers

Our second model takes a representation learning approach, using Doc2Vec to learn dense vector embeddings for stories and various classifiers for comparison.

Model Architecture. We implemented this approach in two stages:

Stage 1: Learning Story Representations. We trained a Doc2Vec model with optimized hyperparameters (450

dimensions, 100 epochs, window size of 10) on the entire corpus of stories. We used 1,039 training examples (39 sample + 1,000 synthetic) and 200 test examples from the development set. Each story (anchor, text A, or text B) was assigned a unique tag and added to the training corpus, resulting in 3,717 tagged documents. The model employed the Distributed Memory (DM) algorithm with mean aggregation, which is better suited for capturing semantic content.

Stage 2: Feature Engineering and Classification. For each triplet, we created a rich 44-dimensional feature vector capturing various similarity and distance metrics between the story embeddings:

- **Basic similarities:** Cosine similarity between anchor and each candidate
- **Distance metrics:** Euclidean, Manhattan, and dot product distances
- **Differential features:** Differences and ratios between candidate similarities
- **Statistical features:** Mean, standard deviation, and norms of embeddings
- **Advanced features:** Angular distances, confidence scores, and cross-similarities

We trained and evaluated five different classifiers using 5-fold cross-validation on the training set, then tested the best performer on the held-out development set.

Results. The Doc2Vec approach demonstrated strong cross-validation performance but revealed significant overfitting when evaluated on the held-out test set.

Cross-Validation Performance (Training Set): All classifiers showed excellent performance during cross-validation:

- **XGBoost:** 97.40% ($\pm 1.00\%$)
- **Random Forest:** 96.63% ($\pm 0.69\%$)
- **Logistic Regression:** 96.53% ($\pm 0.99\%$)
- **MLP Neural Network:** 96.53% ($\pm 1.31\%$)
- **Gradient Boosting:** 96.44% ($\pm 0.90\%$)

Test Set Performance (Development Set): When evaluated on the 200-example held-out development set, performance dropped dramatically:

- **Gradient Boosting:** 55.0% (best test performance)
- **Logistic Regression:** 50.0%
- **MLP Neural Network:** 50.0%
- **XGBoost:** 48.5%
- **Random Forest:** 48.0%

The stark contrast between cross-validation (96-97%) and test set (48-55%) performance indicates severe overfitting. The models learned patterns specific to the synthetic training data that did not generalize to the development set. This suggests that the synthetic training data may have different distributional properties than the human-annotated development data, or that the models were too complex for the limited training set size.

3.3 Sentence Transformer with MLP Classifier

To investigate whether pre-trained embeddings could better capture narrative similarity, we implemented a system using the all-mpnet-base-v2 Sentence Transformer model.

Model Architecture. This approach uses pre-trained embeddings from a robust Sentence Transformer model combined with a neural network classifier:

Stage 1: Encoding Stories. We used the all-mpnet-base-v2 model to encode all stories into 768-dimensional embeddings. This model is pre-trained on a large corpus of sentence pairs and produces normalized embeddings optimized for semantic similarity tasks.

Stage 2: Feature Engineering. Similar to the Doc2Vec approach, we created a comprehensive 49-dimensional feature vector for each triplet, including:

- Cosine similarities and distances between anchor and candidates
- Cross-similarities between the two candidates
- Differential features (ratios, differences, absolute values)
- Statistical and geometric features (norms, angles, means, standard deviations)
- Discriminative features (confidence scores, decision margins)

Stage 3: Neural Network Classification. We trained a Multi-Layer Perceptron (MLP) with multiple hidden layers. We tested four different architectures using 5-fold cross-validation on the training set (539 examples: 39 sample + 1900 synthetic).

Results. The Sentence Transformer approach showed similarly strong cross-validation performance but continued to demonstrate overfitting:

Cross-Validation Performance (Training Set): Different MLP architectures achieved:

- **Medium (128, 64, 32):** 96.66% ($\pm 1.61\%$)
- **Small (64, 32):** 96.29% ($\pm 1.17\%$)
- **Large (150, 75):** 95.92% ($\pm 1.91\%$)
- **Deep (256, 128, 64):** 95.92% ($\pm 1.81\%$)

Test Set Performance (Development Set): The best configuration (Medium architecture) achieved **63.0%** accuracy on the held-out development set of 200 examples. The classification report showed:

- **Text B is closer:** Precision 0.60, Recall 0.76, F1-score 0.67
- **Text A is closer:** Precision 0.68, Recall 0.51, F1-score 0.58

The confusion matrix revealed an imbalance in predictions:

- True Negatives (B is closer, predicted B): 75
- True Positives (A is closer, predicted A): 51
- False Positives (B is closer, predicted A): 24
- False Negatives (A is closer, predicted B): 50

While the Sentence Transformer approach achieved better generalization than Doc2Vec (63% vs 55%), it still showed a significant gap between cross-validation (96.66%) and test performance (63.0%), indicating persistent overfitting. The model showed a bias toward predicting “Text B is closer,” achieving better recall on that class (76%) compared to “Text A is closer” (51%).

4 Track B: Narrative Representation Learning

Track B focuses on learning meaningful vector representations of narratives without explicit supervision on similarity comparisons. The task requires systems to produce embeddings such that cosine similarity between vectors aligns with human judgments of narrative similarity. While Track B evaluation uses individual story embeddings, the organizers validate these embeddings using the labeled triplets from Track A.

4.1 Evaluation Protocol

The evaluation procedure for Track B works as follows:

1. Systems encode all stories from the Track B data into fixed-dimensional vectors.
2. For each triplet in Track A, the system retrieves the embeddings for the anchor, text A, and text B.
3. Cosine similarities are computed: $sim_a = \cos(v_{anchor}, v_a)$ and $sim_b = \cos(v_{anchor}, v_b)$.
4. The system predicts that text A is closer if $sim_a > sim_b$.
5. Accuracy is computed by comparing predictions against the Track A labels.

This evaluation framework tests whether the learned representations capture narrative similarity in a way that generalizes to comparative judgments. Unlike Track A approaches that use rich feature engineering and supervised classifiers, Track B systems rely entirely on the geometric relationships (cosine similarity) between learned embeddings.

4.2 Official Baseline: Sentence-BERT

The organizers provided a Sentence-BERT baseline using the all-MiniLM-L6-v2 model. This lightweight model (only 22.7M parameters) has been pretrained on semantic similarity tasks and produces 384-dimensional embeddings.

The baseline implementation:

- Encodes all Track B stories using the pretrained Sentence-BERT model.
- Creates a lookup dictionary mapping story texts to their embeddings.
- Evaluates on Track A triplets by comparing cosine similarities between anchor-candidate pairs.

Results: The Sentence-BERT baseline achieved an accuracy of **55.0%** on the development set of 200 triplets. While this exceeds the random baseline (50%), the modest improvement suggests that general-purpose semantic similarity embeddings capture only some aspects of narrative similarity. The model may struggle with the nuanced differences between narratives that share semantic content but differ in plot structure, character development, or causal relationships.

4.3 E5-Large-v2 Embeddings

To explore whether a more powerful pretrained model could better capture narrative similarity, we implemented a Track B system using the E5-large-v2 embeddings model.

Model Architecture. E5 (Embeddings from bidirectional Encoder representations) is a family of text embedding models trained with a contrastive learning objective. The `intfloat/e5-large-v2` model:

- Contains significantly more parameters than the Sentence-BERT baseline, resulting in higher-capacity representations
- Produces 1024-dimensional embeddings (compared to SBERT’s 384 dimensions)
- Uses a specialized encoding format with a passage: prefix to optimize retrieval performance
- Produces normalized embeddings optimized for cosine similarity comparisons
- Has been trained on a diverse corpus of text pairs with similarity labels using contrastive learning

Implementation. Our implementation followed a pure embedding approach with no additional training, fine-tuning, or feature engineering:

1. Load the pretrained E5-large-v2 model from Hugging-Face.
2. Encode all 479 Track B stories with the passage: prefix and normalize embeddings.
3. Create an embedding lookup dictionary mapping story texts to their 1024-dimensional vectors.
4. For each of the 200 Track A triplets, retrieve embeddings and compute cosine similarities.
5. Predict that text A is closer if $\cos(v_{anchor}, v_a) > \cos(v_{anchor}, v_b)$.

The system runs entirely on pretrained representations, making predictions based purely on the geometric relationships between story embeddings in the E5 space.

Results. We evaluated the E5-large-v2 model on the development set consisting of 200 Track A triplets. The implementation successfully encoded all 479 Track B stories and performed similarity-based predictions without any missing triplets.

The model achieved an accuracy of **67.0%** on the Track A evaluation triplets. This represents a **12 percentage point improvement** (67.0% vs 55.0%) over the Sentence-BERT baseline, suggesting that E5’s larger capacity (1024 vs 384 dimensions), more sophisticated architecture, and contrastive training objective better capture narrative similarity patterns.

5 Analysis

5.1 Track A: Triplet-based Relative Similarity

Our experiments on Track A reveal several important findings:

Baseline Performance. The GPT-4o-mini baseline’s performance (49.5%) falling below random chance suggests that zero-shot prompting of large language models may not be sufficient for this task. The model may struggle with the specific comparative judgment format or with patterns in the synthetic training data.

Overfitting Challenges. Both our Doc2Vec and Sentence Transformer approaches exhibited severe overfitting, with cross-validation scores around 96-97% but test scores of only 55-63%. This gap indicates that:

1. The synthetic training data may have different distributional properties than the human-annotated development data

2. The models learned superficial patterns specific to the synthetic data generation process
3. Limited training data (500-1,000 examples) is insufficient for the model complexity we employed

Feature Engineering Impact. Despite the rich feature engineering (44-49 dimensional feature spaces capturing various similarity metrics), the models failed to generalize effectively. This suggests that hand-crafted features based on embedding similarities may not capture the deeper narrative properties that distinguish similar from dissimilar stories.

Generalization Gap. The Sentence Transformer approach (63% test accuracy) outperformed Doc2Vec (55% test accuracy), suggesting that pre-trained embeddings capture more generalizable semantic features than task-specific Doc2Vec embeddings trained on limited data. However, both approaches fell short of strong performance, indicating that narrative similarity requires more sophisticated modeling approaches or significantly more training data.

5.2 Track B: Narrative Representation Learning

The E5-large-v2 results demonstrate several important findings:

Embedding Quality Matters: The 12-point improvement over SBERT shows that higher-quality pretrained embeddings can capture more nuanced aspects of narrative similarity. E5’s contrastive training on diverse text pairs appears to create representations that align better with human judgments of narrative closeness.

Pure Embeddings vs. Supervised Models: Interestingly, the pure embedding approach (67%) outperformed our supervised Track A models that used extensive feature engineering. Our Doc2Vec with classifiers achieved only 48-55% test accuracy, and our Sentence Transformer with MLP achieved 63%—both lower than the simple E5 cosine similarity baseline. This suggests that:

- The rich feature engineering in Track A may have introduced noise or overfitting
- Training supervised classifiers on limited synthetic data (500-1,000 examples) may be less effective than leveraging high-quality pretrained embeddings
- Simple cosine similarity on good embeddings can outperform complex supervised pipelines when training data is limited

Remaining Challenges: Despite the improvement, 67% accuracy still leaves substantial room for growth. The model may struggle with:

- Narratives that share semantic content but differ in plot structure or causality

- Fine-grained distinctions between stories with similar themes but different character arcs
- Detecting narrative similarity beyond surface-level semantic overlap

6 Future Work

Our experiments reveal several promising directions for improving narrative similarity systems:

Addressing the Synthetic-to-Real Gap. The severe overfitting observed in Track A (96-97% CV vs. 48-63% test accuracy) highlights the critical challenge of training on synthetic data. Future work should explore:

- **Domain adaptation techniques** to bridge the distributional gap between synthetic training data and human-annotated test data
- **Mixed training strategies** combining synthetic examples with human-annotated data to improve generalization
- **Quality filtering** of synthetic data to retain only examples that align with human judgments
- **Adversarial validation** to identify and exclude synthetic examples that are too dissimilar from real test distributions

Fine-tuning Pretrained Embeddings. Given that E5-large-v2 achieved the best performance (67%) using zero-shot embeddings, fine-tuning on narrative-specific data could yield substantial improvements:

- Fine-tune E5 or similar models on narrative similarity pairs using contrastive learning objectives
- Leverage the human-annotated development data for few-shot adaptation
- Explore multi-task learning combining narrative similarity with related tasks like plot structure analysis or character relationship extraction

Narrative-Specific Features. Pure semantic embeddings may miss structural narrative properties. Future systems could incorporate:

- **Character networks** to capture similarities in character relationships and interactions
- **Event sequence modeling** to compare plot structures and narrative arcs
- **Causal chain extraction** to identify similarities in cause-effect relationships
- **Temporal structure analysis** to capture pacing and narrative progression patterns

Ensemble Approaches. Combining multiple embedding models and approaches could leverage their complementary strengths:

- Ensemble E5, Sentence Transformers, and fine-tuned models
- Combine embedding-based approaches with symbolic methods that explicitly model narrative structure
- Use stacking or meta-learning to optimally weight different models' predictions

Data Augmentation Strategies. While we experimented with back-translation in our preliminary work, several augmentation techniques warrant exploration that preserve narrative structure while increasing surface-level diversity:

- Synonym replacement and lexical substitution to vary word choice without altering story semantics
- LLM-based paraphrasing with controlled generation to rewrite stories while maintaining plot structure and causal relationships
- Story compression and expansion to test robustness to different narrative granularities
- Hard negative mining through systematic perturbations (character role swaps, ending modifications) to create challenging distinction tasks
- Counterfactual generation and genre transfer to explore narrative variations while preserving core story elements

7 Conclusion

This report presents our exploration of narrative similarity for SemEval-2026 Task 4, implementing Doc2Vec with multiple classifiers, Sentence Transformer with MLP, and E5-large-v2 embeddings for Track A (triplet-based similarity) and Track B (representation learning).

Our experiments revealed that E5-large-v2 with pure cosine similarity achieved the best performance at 67% accuracy, outperforming complex supervised methods (Doc2Vec: 48-55%, Sentence Transformer + MLP: 63%). The Track A models exhibited severe overfitting (96-97% cross-validation vs. 48-63% test accuracy), highlighting a critical synthetic-to-real generalization gap. The GPT-4o-mini baseline performed below random chance (49.5%), suggesting zero-shot prompting is insufficient for this task.

These results demonstrate that high-quality pretrained embeddings outperform complex feature engineering when training data is limited. Future work should focus on fine-tuning embeddings on narrative-specific data, addressing the synthetic-to-real gap through improved augmentation strategies, and incorporating explicit structural features like character networks and event sequences to capture deeper narrative properties beyond surface-level semantics.